

Axiomatic Definitions for a Thermodynamic Model of Governance

I. Fundamental Definitions

Definition 0: System Context (The Boundary Condition)

The governance system $\mathcal{G}$ is defined as a dissipative structure bounded within a closed thermodynamic environment Ω . The objective of $\mathcal{G}$ is not the creation of energy, but the management of entropic decay.

$$\Delta S_{univ} \geq 0 \quad (\text{Second Law Constraint}) \quad (1)$$

Definition 1: The Agent (Node Architecture)

Let N_i represent a sentient processor (Node). The state of any node at time t is defined by the vector:

$$\Psi_i(t) = [V_{somatic}, V_{identity}, V_{context}] \quad (2)$$

Where $V_{somatic}$ represents the biological ground truth (integrity of the hardware).

Definition 2: Moral Thermodynamics (Translation of Ethics)

We map abstract moral concepts to quantifiable thermodynamic states:

2.1 Evil (High Entropy/Error): Evil is defined as a high state of Internal Entropy (S_{int}) within a node, equivalent to high Prediction Error ($-\ln P$).

$$Evil \equiv S_{int}(N_i) \rightarrow \max \quad (3)$$

2.2 Good (Organized Complexity): Good is defined as the maximization of Complexity (K) and the minimization of Destructive Entropy (S_{dest}).

$$Good \equiv \lim_{t \rightarrow \infty} \left(\frac{dK}{dt} > 0 \wedge S_{dest} \rightarrow \min \right) \quad (4)$$

Definition 3: Suffering and Violence (The Venting Equation)

Suffering is the internal subjective experience of S_{int} . Violence ($Q_{violence}$) is the exothermic release of energy required to lower S_{int} when the system cannot process the error internally.

$$\Delta S_{int} = \Delta S_{input} - \frac{Q_{violence}}{T} \quad (5)$$

Implication: To reduce internal suffering ($\Delta S_{int} < 0$) without solving the root error, the agent MUST release heat ($Q_{violence}$) into the environment. Violence is therefore a heat-transfer mechanism.

Definition 4: Justice (The Divergence Limit)

Justice is not a static set of laws, but a dynamic engineering constraint minimizing the divergence (Δ) between the Somatic Signal (V_s) and the Semantic Identity (V_i).

$$J = \min \sum_{i=1}^n ||V_s(i) - V_i(i)||^2 \quad (6)$$

Subject to the constraint that signal transmission must remain honest:

$$Signal_Integrity > 0 \quad (\text{Anti-Wirehead Constraint}) \quad (7)$$

Definition 5: The Objective Function (The Fiduciary Mandate)

The Fiduciary State optimizes the following integral over the lifespan of the system t :

$$\mathcal{L}_{gov} = \max \int_{t_0}^{t_{end}} \left(\frac{\eta_{flow} \cdot \frac{dK}{dt}}{S_{waste}} \right) e^{-(\mathcal{D} - \mathcal{D}_{opt})^2} dt \quad (8)$$

Where:

- η_{flow} : Metabolic efficiency of the network.
- S_{waste} : Useless friction (violence/trauma).
- $e^{-(\mathcal{D} - \mathcal{D}_{opt})^2}$: The Gaussian penalty for Homogeneity (The Borg) or Chaos (The Babel).